

Scope sensitivity: 30 retained tasks

A separate copy of the tier-colored metric report after removing every task from six papers. The underlying runs are unchanged. This exclusion was chosen after observing tier fit, so this is an exploratory sensitivity analysis, not an independent or prospectively selected benchmark.

T1 (expected moderate use) T2 (expected strongest use) T3 (expected little/no use)

Task-level medians in the retained cohort

Tier	Tasks	Distinct names	Occurrences	Audited deps	Time gain
T1	9	6.0	8.0	22.7	+62.5%
T2	3	11.0	18.3	95.3	+75.3%
T3	18	0.0	0.0	0.0	+6.3%

Interpretation limit: the excluded papers were selected after results were inspected; the apparent association can improve or deteriorate simply because of selection. T1/T2 and T3 also came from different runner campaigns. P15-T3, the strongest low-expected-use exception, remains in this cohort.

Retained: 30 of 38 tasks, 90 of 114 paired runs. Excluded papers: P02, P17, P20, P33, P35, P40. The original 38-task report and all raw evidence remain preserved. Positive gain means L used less than N.

What changed when six papers were removed

The eight removed tasks are listed below by paper. All three P15 tasks, including P15-T3, remain. No formalization or proof was rerun for this comparison.

P02: P02-T1, P02-T3	P17: P17-T1, P17-T3	P20: P20-T3
P33: P33-T1	P35: P35-T3	P40: P40-T3

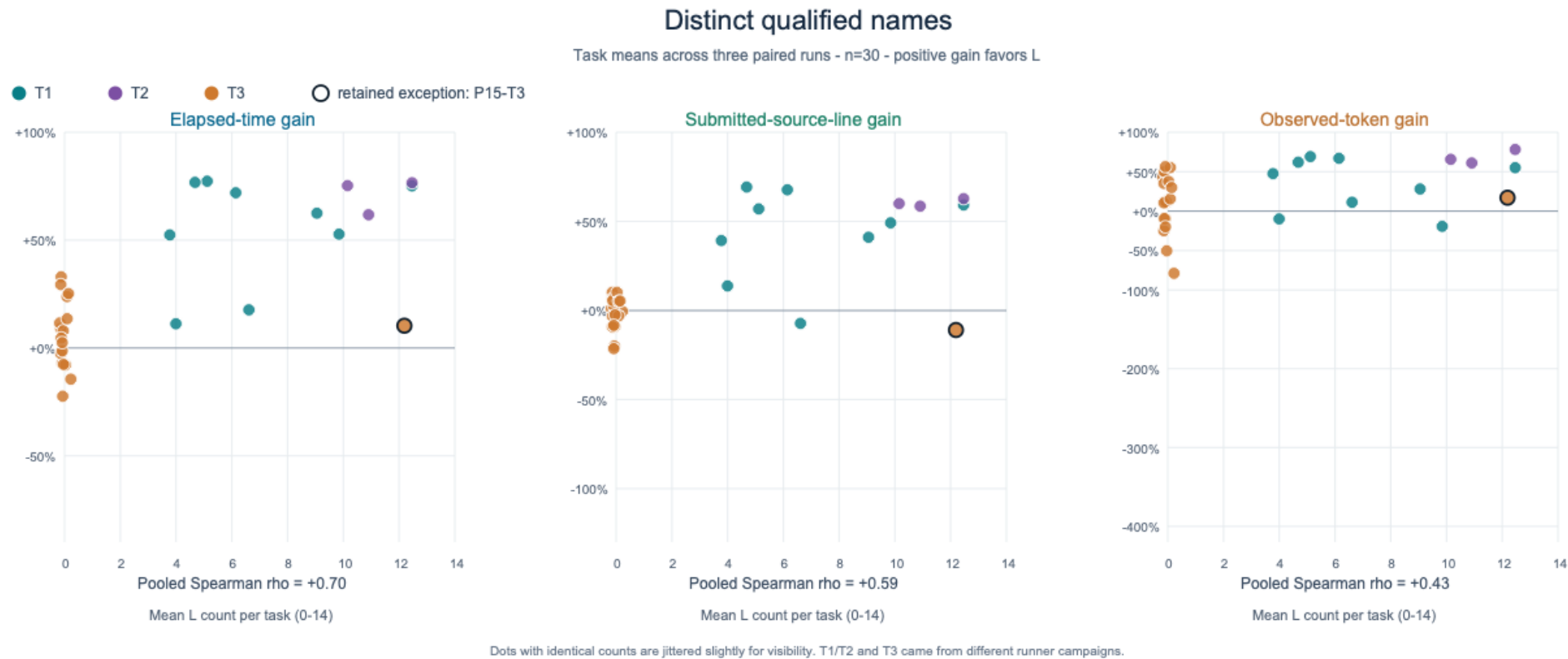
Overall median gain, full to retained: time +13.9% to +15.6%; lines +4.9% to +5.4%; tokens +13.7% to +29.0%.

Pooled Spearman rho: full 38 versus retained 30

Usage measure	Gain	Full	Retained	Change
Distinct qualified names	Time	+0.64	+0.70	+0.07
Distinct qualified names	Lines	+0.49	+0.59	+0.10
Distinct qualified names	Tokens	+0.28	+0.43	+0.15
Qualified-name occurrences	Time	+0.58	+0.69	+0.11
Qualified-name occurrences	Lines	+0.44	+0.58	+0.15
Qualified-name occurrences	Tokens	+0.20	+0.40	+0.20
Audited transitive deps	Time	+0.67	+0.73	+0.06
Audited transitive deps	Lines	+0.57	+0.66	+0.10
Audited transitive deps	Tokens	+0.33	+0.47	+0.14

These coefficients are descriptive. A larger rho after filtering does not validate the post-run exclusion or establish that the excluded mathematics was out of scope. The full 12-measure comparison is in `association_change.csv`.

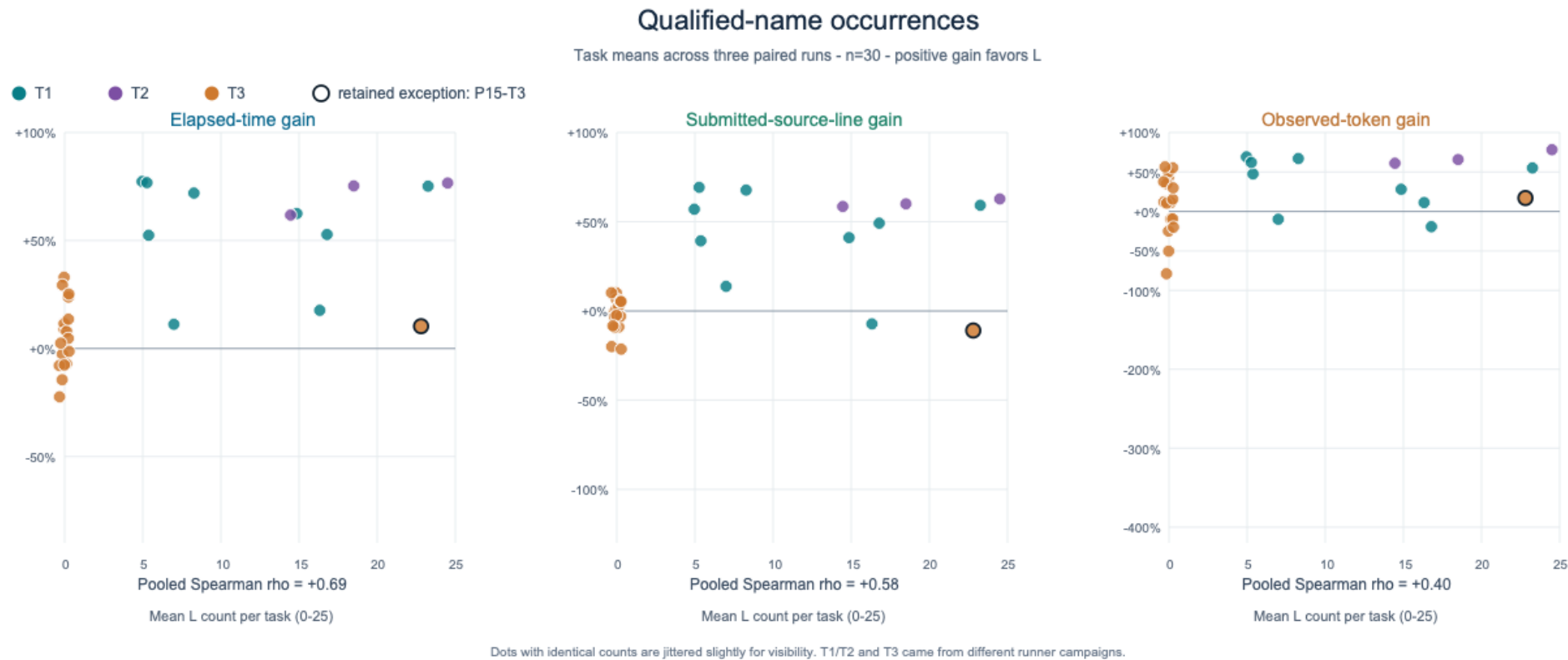
Metric 1/12: Distinct qualified names



Definition: Distinct NumStability.* names written outside imports, comments and strings.

This plot uses a post-run filtered cohort. Tier and runner campaign remain confounded; the pooled correlation is descriptive, not causal.

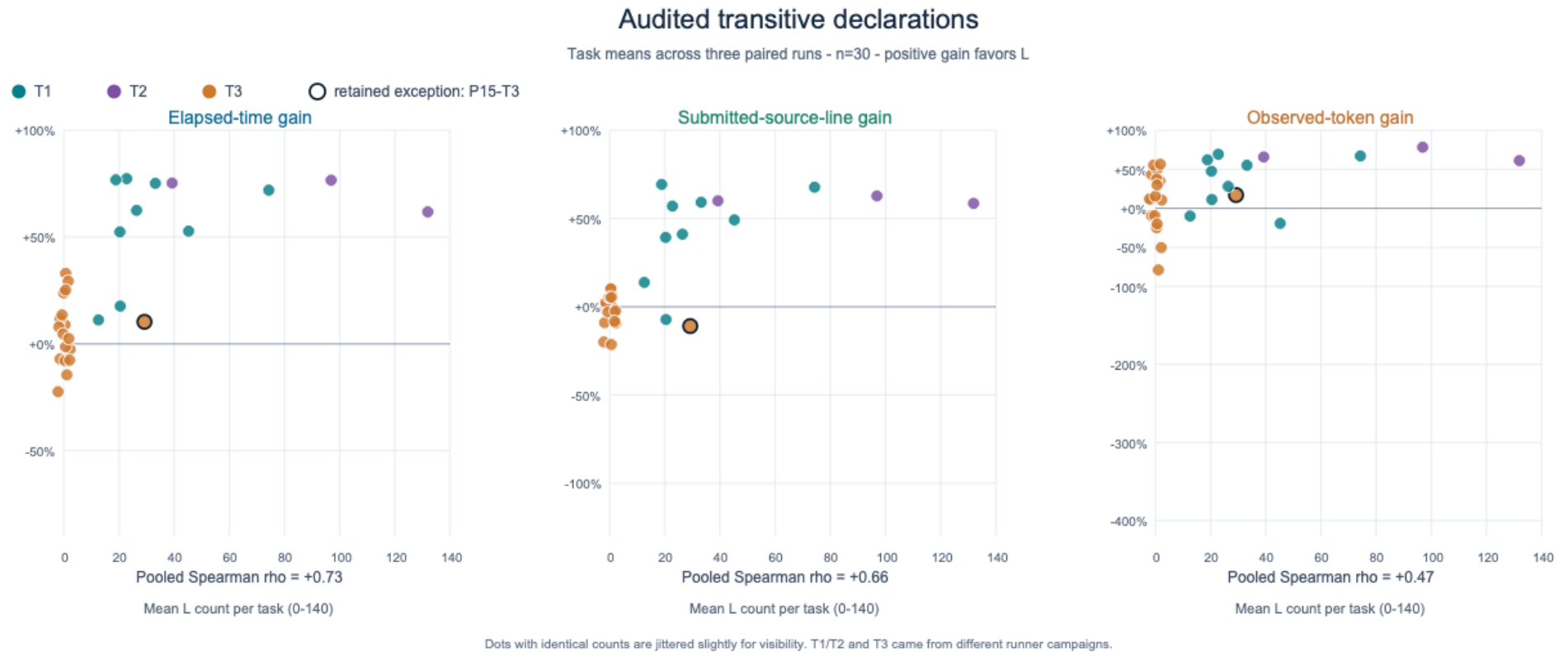
Metric 2/12: Qualified-name occurrences



Definition: Every NumStability.* occurrence written outside imports, comments and strings.

This plot uses a post-run filtered cohort. Tier and runner campaign remain confounded; the pooled correlation is descriptive, not causal.

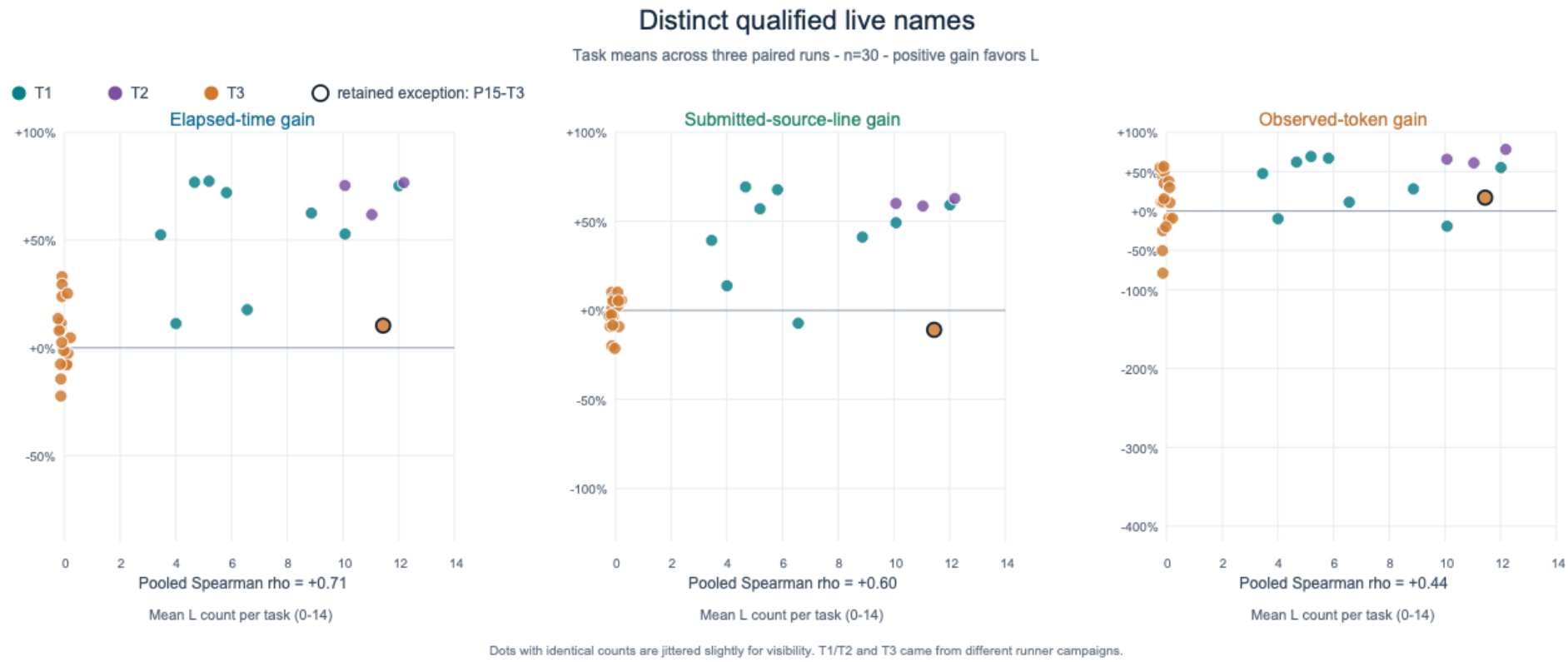
Metric 3/12: Audited transitive declarations



Definition: All NumStability dependencies in the accepted theorem's audited dependency closure.

This plot uses a post-run filtered cohort. Tier and runner campaign remain confounded; the pooled correlation is descriptive, not causal.

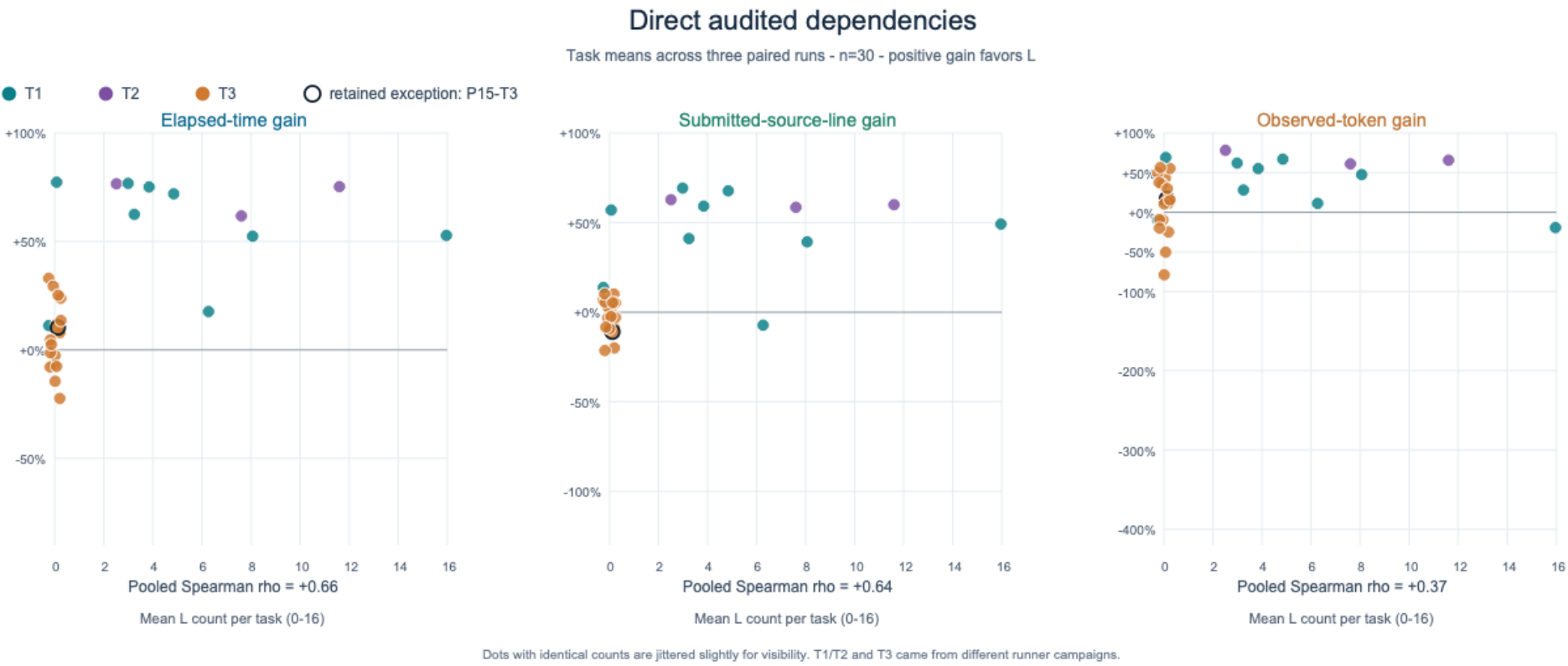
Metric 4/12: Distinct qualified live names



Definition: Written qualified names also matched to a name in the audited closure.

This plot uses a post-run filtered cohort. Tier and runner campaign remain confounded; the pooled correlation is descriptive, not causal.

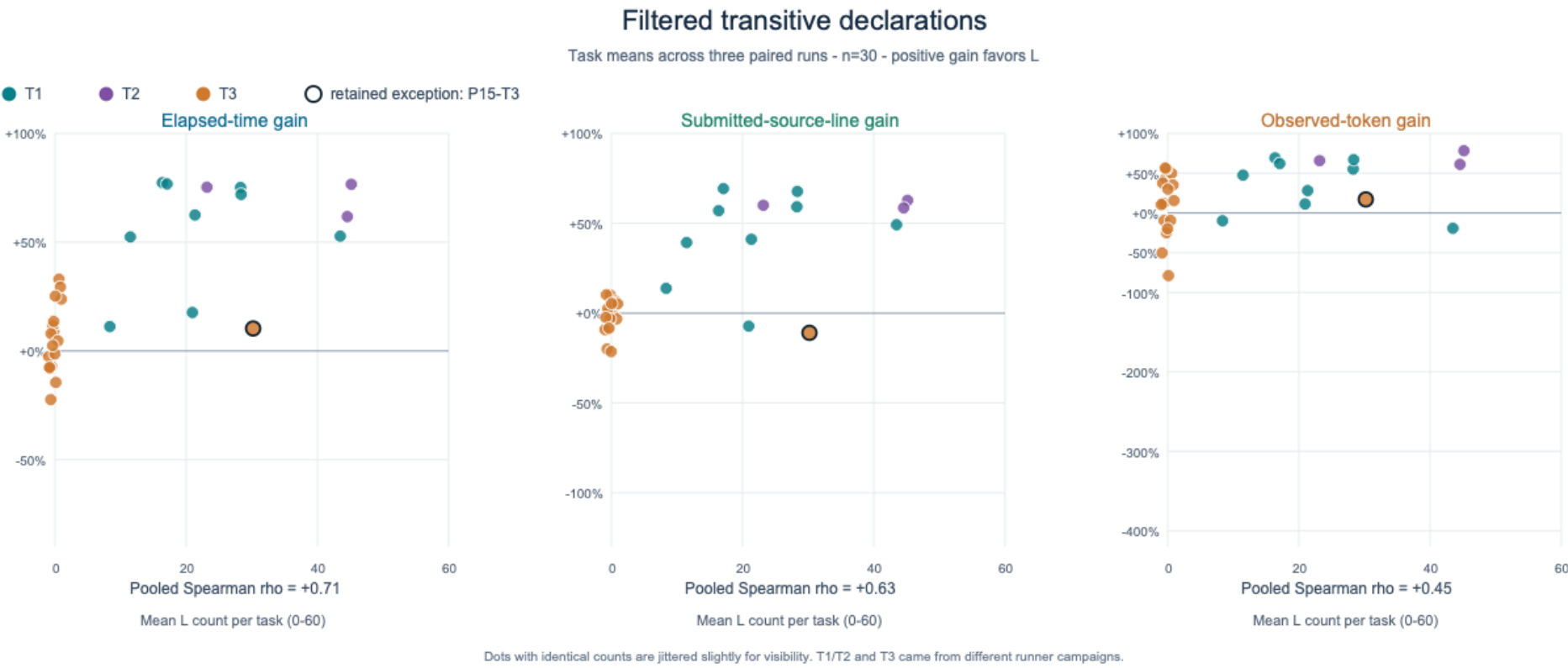
Metric 5/12: Direct audited dependencies



Definition: NumStability declarations at dependency-graph distance one from the accepted theorem.

This plot uses a post-run filtered cohort. Tier and runner campaign remain confounded; the pooled correlation is descriptive, not causal.

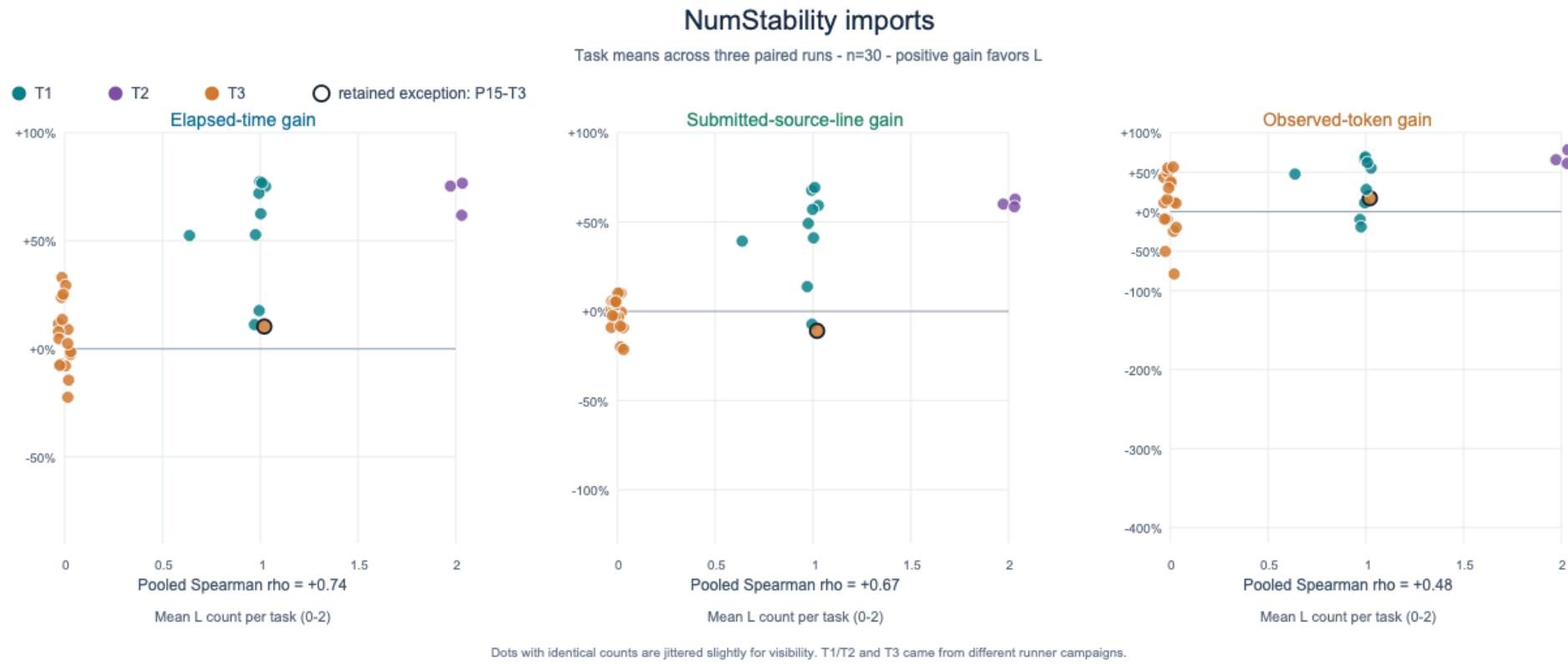
Metric 6/12: Filtered transitive declarations



Definition: Transitive declarations after the archive's limited generated-name filter.

This plot uses a post-run filtered cohort. Tier and runner campaign remain confounded; the pooled correlation is descriptive, not causal.

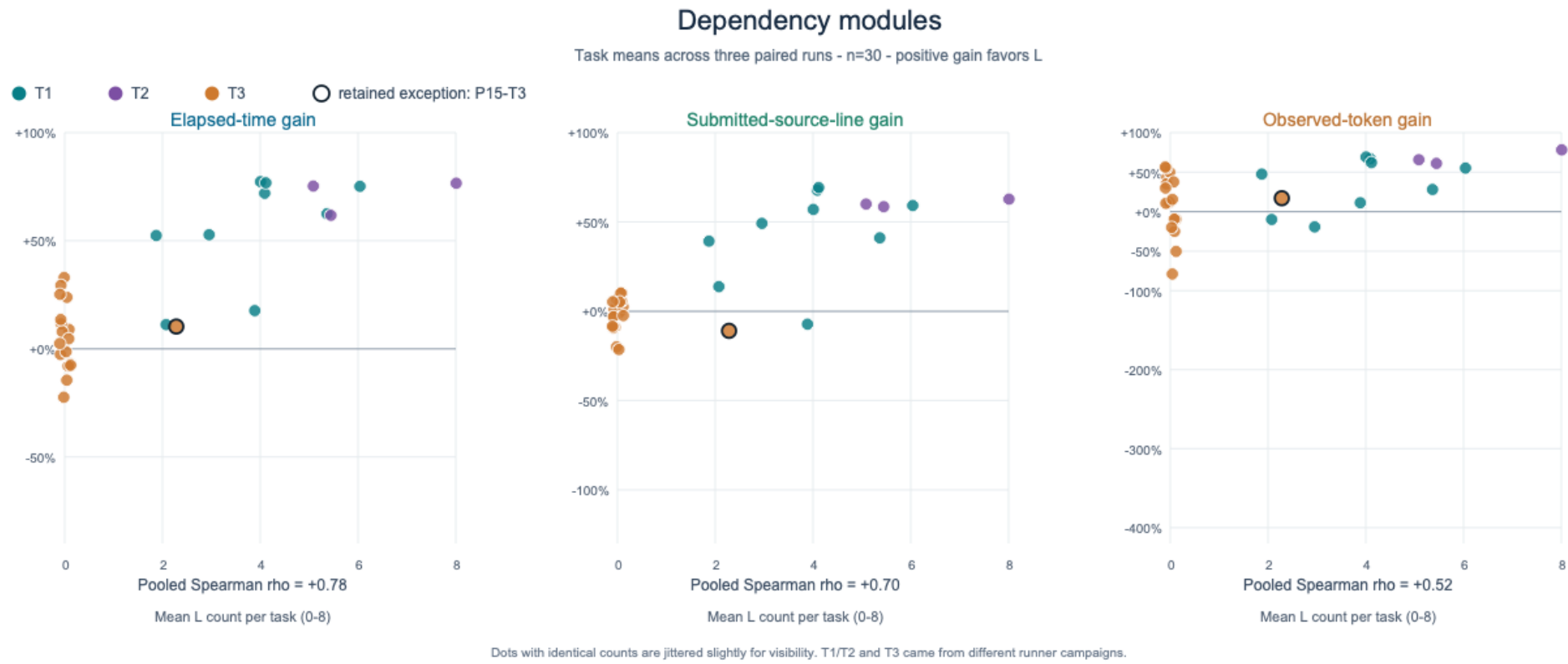
Metric 7/12: NumStability imports



Definition: Distinct NumStability import lines in the accepted source.

This plot uses a post-run filtered cohort. Tier and runner campaign remain confounded; the pooled correlation is descriptive, not causal.

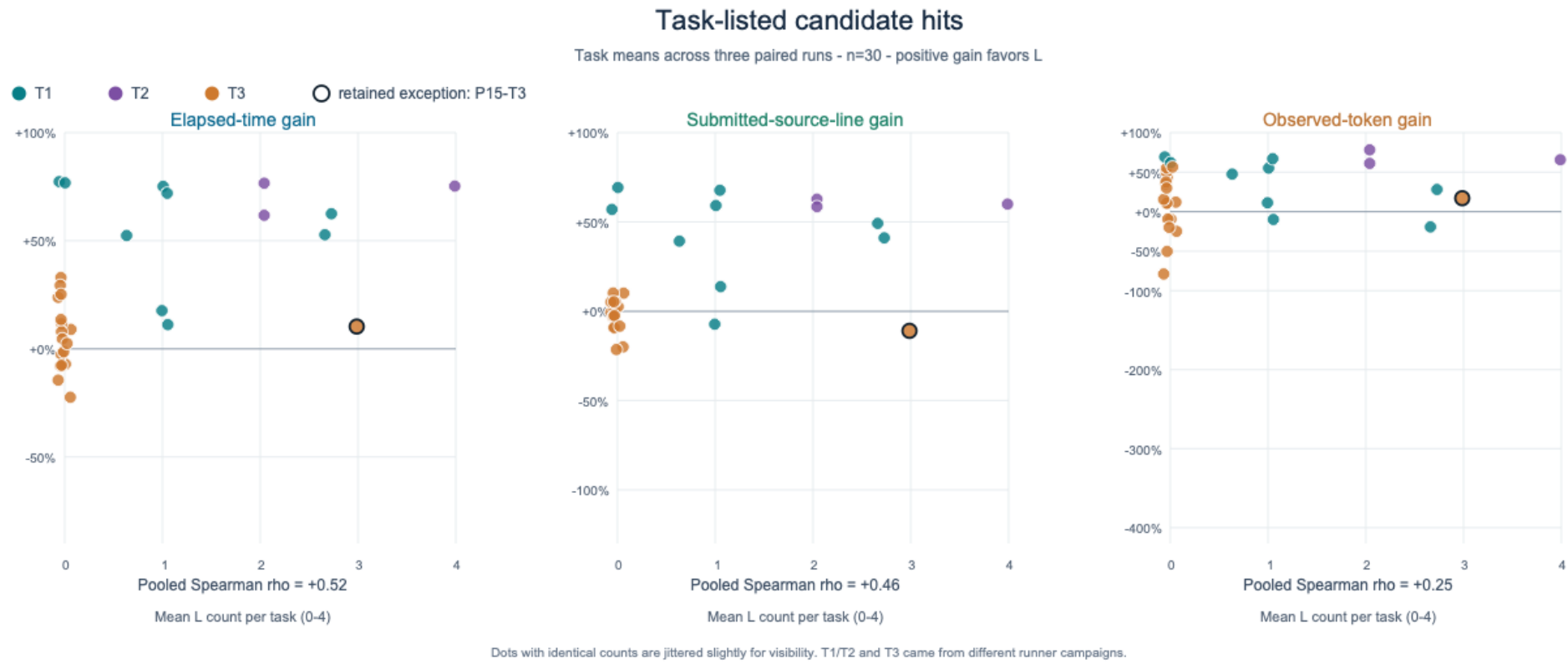
Metric 8/12: Dependency modules



Definition: Distinct NumStability modules owning audited dependency declarations.

This plot uses a post-run filtered cohort. Tier and runner campaign remain confounded; the pooled correlation is descriptive, not causal.

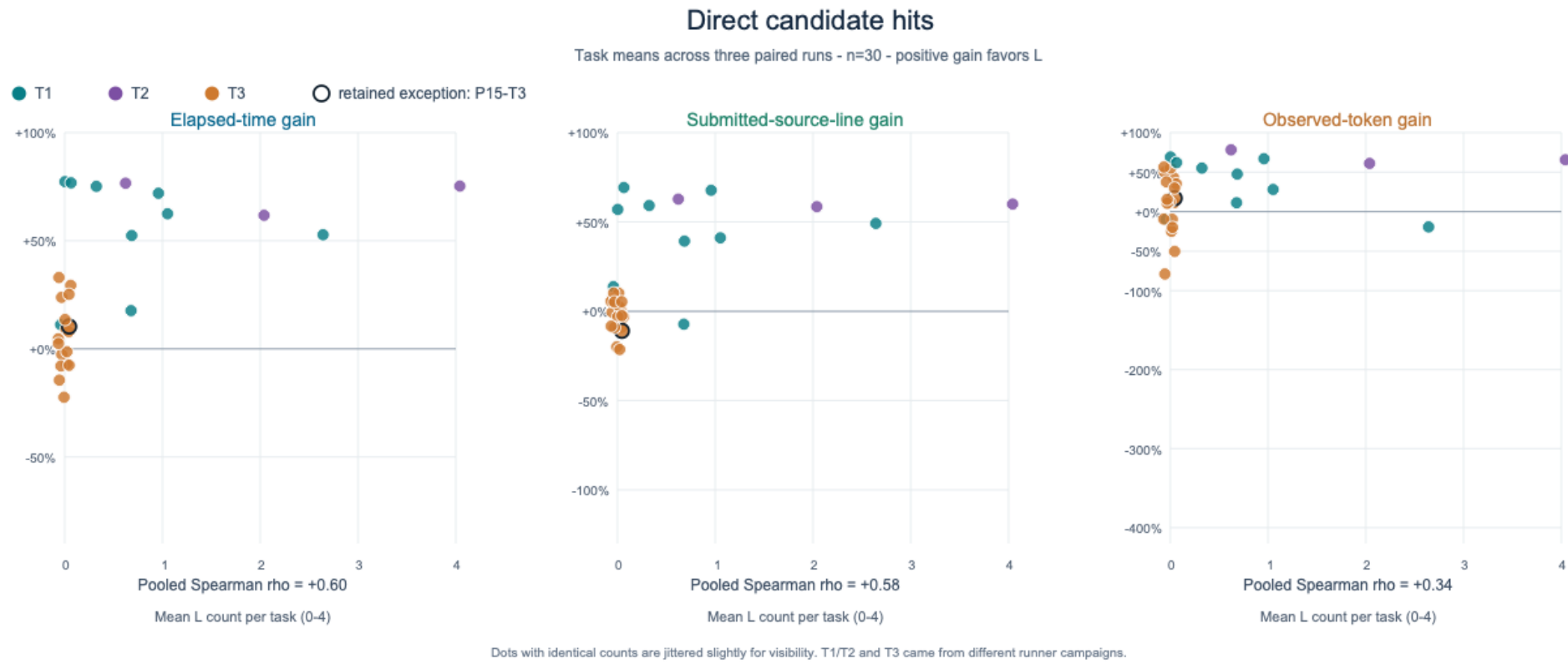
Metric 9/12: Task-listed candidate hits



Definition: Task-specific candidate names present anywhere in the audited dependency closure.

Task-specific candidate classifications enter this measure. T3 has no predesignated substantial result, so substantial-hit zeros there are partly by construction.

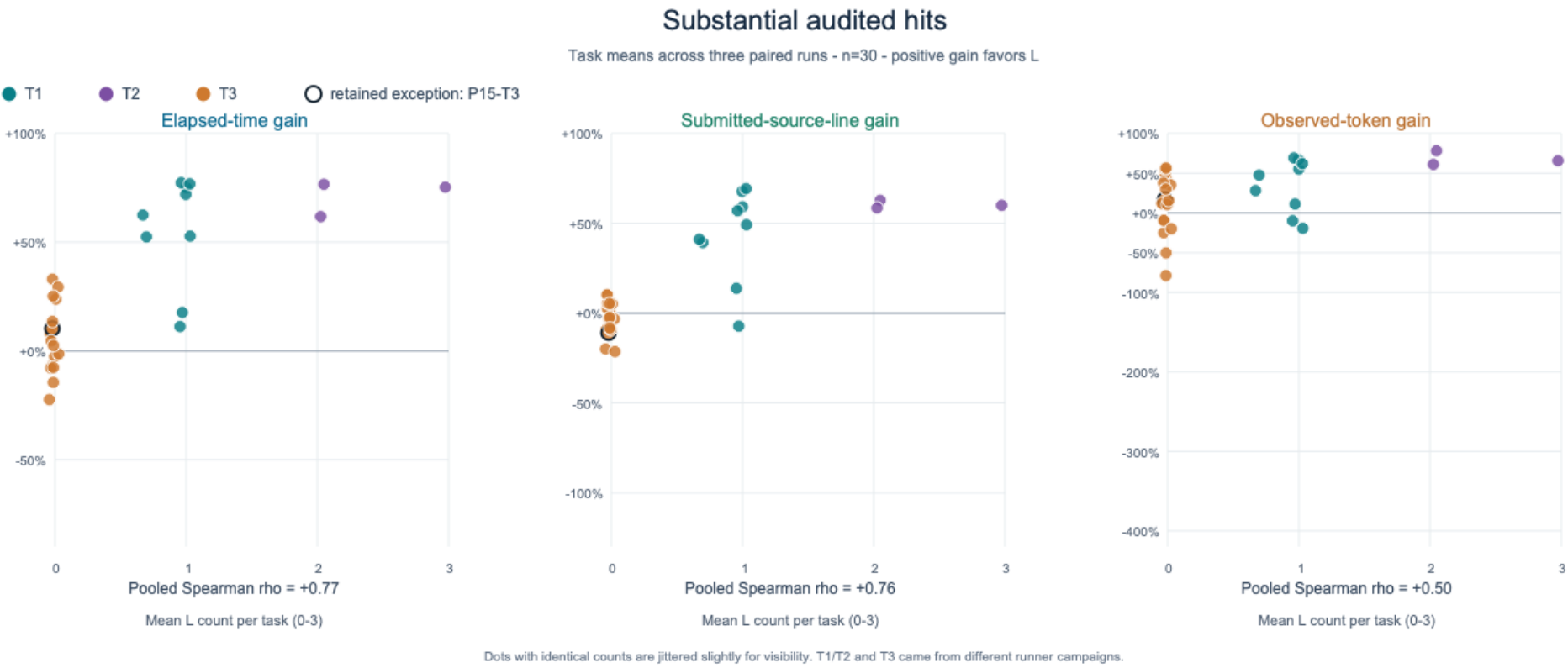
Metric 10/12: Direct candidate hits



Definition: Task-specific candidate names at dependency-graph distance one.

Task-specific candidate classifications enter this measure. T3 has no predesignated substantial result, so substantial-hit zeros there are partly by construction.

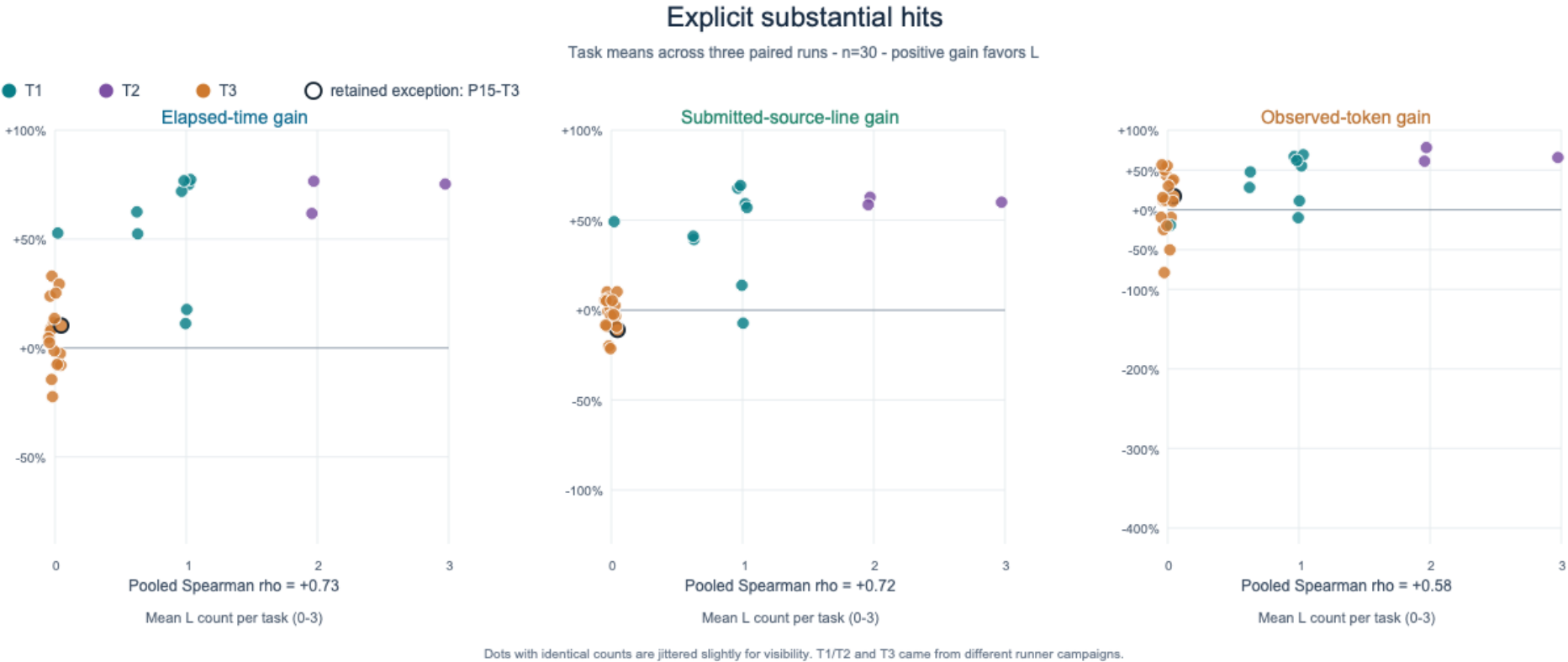
Metric 11/12: Substantial audited hits



Definition: Task-relevant substantial declarations in the audited closure.

Task-specific candidate classifications enter this measure. T3 has no predesignated substantial result, so substantial-hit zeros there are partly by construction.

Metric 12/12: Explicit substantial hits



Definition: Task-relevant substantial declarations named directly in source.

Task-specific candidate classifications enter this measure. T3 has no predesignated substantial result, so substantial-hit zeros there are partly by construction.

Source and scope notes

Original evidence: `paper_benchmark/highambench/reports/library-usage-analysis/attempt_usage.csv`, `task_usage.csv`, `pair_usage.csv`, `analyze.py` and `semantic_catalog.json`; `paper_benchmark/highambench/metadata/manifest.json`. The original tier-colored PDF remains at `reports/metric-comparison/tier_metric_comparison.pdf`, pinned by SHA-256 in `scope_manifest.json`.

Scope rule: remove every task whose paper ID is one of P02, P17, P20, P33, P35, P40. The script asserts the exact eight excluded task IDs and retains P15-T3. There is no task-level outcome filter beyond that paper rule. Every retained task contributes all three original paired runs.

This filtering decision was made after the tier-fit analysis. It changes the analyzed cohort and cannot be presented as the original benchmark corpus or as held-out confirmation. Unexpected tier fit is a diagnostic signal, not by itself proof that a paper is mathematically outside NumStability's aims.

Aggregation and metrics match the preserved report: L-side usage is averaged across three accepted runs per task; gains compare N and L three-run means. Qualified names are lexical; audited dependencies come from the accepted theorem's recorded Lean dependency closure. Transitive counts can include indirect and compiler-generated declarations. Token counts are provider-observed.

Rebuild: run `python3 make_report.py` from this directory. It regenerates 12 tier-colored SVG/PNG figures, `retained_tasks.csv`, `association_change.csv` and this PDF. The script hash-checks the preserved 38-task PDF and imports the source-CSV hash-checked analysis functions.